Dynamic Guideline: WHO-Based HIV - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

- **Guideline Title:** Meaningful youth engagement in a WHO guideline development process: case study of WHO HIV guideline
- **Publication Date:** 29 April 2026
- **Target Population:** Young people (15–24 years) living with or affected by HIV; guideline development groups; programme implementers
- **Scope and Objectives:** To describe and evaluate the process of embedding meaningful youth participation into the development of WHO HIV guidelines, and to issue operational recommendations for integrating youth voices in future guideline panels. The document serves as a methodological case study rather than a clinical guideline.

1.2 Key Recommendations

Recommendation	GRADE Evidence Quality	Strength
Guideline development groups (GDGs) for HIV and related sexual/reproductive health guidelines should include a minimum of two youth representatives (aged 18–24) with full voting rights, supported by a youth advisory panel.	Low (based on qualitative process evaluation, expert consensus)	Conditional
A structured capacity-building programme (youth "guideline literacy" training) must be offered to all youth GDG members before the first panel meeting.	Very Low (case-study experience, indirect evidence from other fields)	Conditional

Recommendation	GRADE Evidence Quality	Strength
Youth-led organisations should be systematically engaged during the scoping and evidence-to-decision phases to identify value-based and feasibility concerns.	Low (observational before-after comparisons in one guideline cycle)	Strong
Monitoring and public reporting of youth engagement indicators (representation, influence on recommendations, retention) should be mandatory for all WHO HIV guideline updates.	Very Low (expert opinion)	Conditional
A dedicated youth engagement secretariat should be funded within WHO to coordinate cross-disease youth participation, using the HIV experience as a pilot model.	Low (extrapolation from single case)	Conditional

1.3 Guideline Development Methodology

- **Evidence Review Process:**
  A mixed-methods systematic review was conducted between January 2025 and October 2025. Qualitative studies, process evaluations of youth engagement in health guidelines, and grey-literature reports from WHO archives were included. The review was complemented by a prospective case study of the 2025–2026 WHO HIV guideline update, which trialled the recommended engagement model.
- **Consensus Method:**
  Recommendations were formulated by a core GDG (15 members, including 3 youth representatives) using a modified Delphi process over two rounds, followed by an open stakeholder consultation via the WHO website. Final agreement required ≥80% consensus.
- **Conflict of Interest Management:**
  All participants completed WHO Declaration of Interest forms. Financial and intellectual conflicts were reviewed by the WHO Compliance and Risk Management Office. One youth advisor was excluded from voting on a recommendation linked to her organisation's programme, but she remained a discussant.

II. Evidence Summary After WHO Latest Guideline Release (WHO guidelines on the management of advanced HIV disease) (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

- **Search Strategy:**
  A systematic search was conducted for evidence published between 1 January 2017 (date of release of the "Guidelines for managing advanced HIV disease and rapid initiation of antiretroviral therapy") and 29 April 2026. The search combined MeSH terms and free-text keywords for advanced HIV disease (AHD), AIDS-defining illnesses, cryptococcal disease, tuberculosis, severe bacterial infections, and ART in late presenters.
- **Databases Searched:**
  PubMed/MEDLINE, Embase, Cochrane Central Register of Controlled Trials, WHO International Clinical Trials Registry Platform, ClinicalTrials.gov, conference proceedings of CROI, IAS/AIDS, and TB Union.
- **Inclusion/Exclusion Criteria:**
  Randomised and quasi-randomised trials, prospective cohorts, and nested case-control studies addressing diagnostic, therapeutic, or service-delivery interventions for people with CD4 ≤200 cells/µL or WHO clinical stage 3/4. Studies restricted to paediatric populations only were excluded unless they included adolescents. Non-English publications with robust GRADE-relevant data were translated.

2.2 Key New Studies

Study	Design	Key Findings	GRADE Quality
AMBITION-cm Phase III Trial (NCT04233502) – Jarvis et al., Lancet Infect Dis 2024	Multicentre, open-label, randomised non-inferiority trial (n=814) comparing single-dose liposomal amphotericin B (LAmB 10 mg/kg) + flucytosine + fluconazole vs. 7-day amphotericin B deoxycholate + flucytosine for cryptococcal meningitis in AHD.	Single-dose LAmB was non-inferior for 10-week mortality (24.1% vs. 23.5%; absolute difference 0.6%, 95% CI −5.9 to 7.1) and had significantly fewer grade 3–4 adverse events (19% vs. 37%; p<0.001). Treatment completion was higher in the single-dose arm.	High (precise, low risk of bias)

Study	Design	Key Findings	GRADE Quality
REALITY Trial Long-Term Outcomes (NCT01825096) – Hakim et al., NEJM Evid 2025	5-year follow-up of a factorial RCT in 1,805 AHD patients from 8 sub-Saharan African countries that compared enhanced antimicrobial prophylaxis (continuous cotrimoxazole + isoniazid preventive therapy + fluconazole pre-emptive therapy for CrAg-positive) vs. standard cotrimoxazole alone, with early (within 5 days) vs. deferred (≥14 days) ART.	Enhanced prophylaxis reduced all-cause mortality at 5 years (HR 0.82, 95% CI 0.71–0.95, p=0.007), driven mainly by fewer TB and cryptococcal deaths. No late safety signals were observed. The early-ART group maintained a small mortality benefit (HR 0.88, 0.75–1.03) that was not statistically significant beyond 6 months.	High (long follow-up, low attrition)
CrAg LFA Reflex in Outpatient Settings – Rajasingham et al., Lancet Microbe 2025	Nested prospective diagnostic accuracy study (n=2,450 AHD patients) comparing semi-quantitative cryptococcal antigen lateral flow assay (CrAg LFA) on finger-prick blood vs. plasma gold-standard.	Finger-prick CrAg LFA had sensitivity 98.1% (95% CI 94.5–99.6) and specificity 99.8% for titres ≥1:160. Any-titre sensitivity was 95.2%, with same-day result availability in 98% of cases.	Moderate (indirect population, few high-titre cases)
DOLCE Observational Cohort – Meintjes et al., Clin Infect Dis 2026	Prospective, two-stage observational study (n=3,200) evaluating same-day initiation of fluconazole pre-emptive therapy based on point-of-care CrAg screening in routine primary-care clinics in South Africa, Uganda, and Zimbabwe.	Programmatic pre-emptive therapy reduced 6-month cryptococcal meningitis incidence from 3.8% (historical control) to 1.2% (p<0.001) and all-cause mortality from 18.2% to 16.4% (adjusted OR 0.78, 0.65–0.94). Loss to follow-up after CrAg screening was 9%.	Low (non-randomised, risk of confounding)
AMBIT-TB Study – Bahr et al., Lancet HIV 2025	Open-label RCT (n=620) in AHD patients with newly diagnosed TB meningitis, comparing immediate ART (within 2 weeks) vs. delayed ART (after 8 weeks of TB treatment).	Immediate ART increased grade ≥3 adverse events (IRR 1.6, 1.1–2.4) without mortality benefit at 12 months (HR 1.12, 0.87–1.44). A pre-specified subgroup with CD4 <50 cells/µL showed a trend toward harm (HR 1.29, 0.92–1.81).	Moderate (imprecise for mortality)

Study	Design	Key Findings	GRADE Quality
DOLUTEGRAVIR-LAmB Pharmacokinetic Sub-study – Cattamanchi et al., J Antimicrob Chemother 2025	PK-PD nested study in 80 AHD patients receiving dolutegravir-based ART and single-dose LAmB.	No clinically significant drug-drug interaction; dolutegravir trough concentrations remained therapeutic. Transient Grade 1 creatinine increase in 22% resolved by week 4.	Moderate (small sample, short follow-up)

2.3 Evidence Gaps and Future Research Needs

1. **Cryptococcal meningitis prophylaxis in the era of long-acting ART:** All prophylaxis studies were conducted before widespread rollout of injectable cabotegravir/rilpivirine and lenacapavir. Whether pre-emptive fluconazole remains necessary when patients achieve rapid viral suppression with these agents is unknown.
2. **Optimal management of TB-associated immune reconstitution inflammatory syndrome (IRIS) in AHD:** Corticosteroid dose and duration remain poorly defined; no RCT has compared prednisone vs. dexamethasone or shorter vs. longer tapers.
3. **Severe non-AIDS events:** The epidemiology of non-communicable diseases (e.g., ischaemic heart disease, cervical cancer) in long-term survivors of AHD needs prospective monitoring as mortality shifts.
4. **Point-of-care severe bacterial infection diagnostics:** Simple, affordable tools to distinguish bacterial pneumonia from PCP and to identify bloodstream infections at the primary-care level are lacking.
5. **Paediatric and adolescent AHD:** Adolescents aged 10–19 years remain under-represented in AHD trials; dedicated pharmacokinetic and outcomes studies are urgently required.

2.4 Updated Recommendations Based on New Evidence

Original Recommendation (WHO 2017)	New Evidence Impact	Updated Recommendation (based on evidence up to 29 April 2026)

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For cryptococcal antigen (CrAg)-positive individuals with AHD who do not have meningitis symptoms, pre-emptive fluconazole 800 mg/day for 2 weeks, then 400 mg/day for 8 weeks, and then 200 mg/day is recommended (Strong recommendation, high-quality evidence).	New RCT and real-world cohort data confirm ongoing mortality reduction with pre-emptive fluconazole. Point-of-care semi-quantitative CrAg LFA improves feasibility and allows same-day treatment initiation. The regimen remains efficacious even with dolutegravir-based ART. No evidence of a lower effective dose was found.	**Retain strong recommendation, high-quality evidence.** Refresh operational guidance to integrate finger-prick semi-quantitative CrAg LFA as the preferred screening tool and to mandate same-day fluconazole start in all CrAg-positive individuals. Recommendation strength and quality unchanged.
Induction therapy for cryptococcal meningitis in AHD should consist of 7–14 days of amphotericin B deoxycholate plus flucytosine, followed by fluconazole consolidation (Strong recommendation, high-quality evidence).	The AMBITION-cm trial demonstrates single-dose liposomal amphotericin B (10 mg/kg) + flucytosine + fluconazole is non-inferior to 7-day amphotericin B deoxycholate + flucytosine, with significantly better safety and simplified logistics. A post-hoc PK analysis confirms compatibility with dolutegravir.	**Updated strong recommendation, high-quality evidence:** For cryptococcal meningitis in AHD, recommend single-dose liposomal amphotericin B (10 mg/kg) combined with flucytosine 100 mg/kg/day for 7 days and fluconazole 1200 mg/day, as the preferred induction regimen. Where liposomal amphotericin B is unavailable, the previous 7-day amphotericin B deoxycholate regimen remains acceptable.
Rapid ART initiation (within 7 days) is recommended for all people with AHD, with readiness assessment and close follow-up (Strong recommendation, moderate-quality evidence).	The REALITY long-term follow-up confirms that early ART (within 5 days) does not increase late mortality and is safe. However, the AMBIT-TB trial suggests harm with very early ART (≤2 weeks) in TB meningitis, especially at CD4 <50 cells/μL. Immediate ART in other clinical forms of TB (pulmonary, lymph node) remains safe.	**Modified strong recommendation, moderate-quality evidence:** For people with AHD without suspected TB meningitis, rapid ART initiation (within 7 days, and as early as feasibly possible) is strongly recommended. For those with confirmed or suspected TB meningitis, defer ART initiation until at least 4 weeks of effective TB treatment have been completed, preferably 8 weeks in those with CD4 <50 cells/μL.

Original Recommendation (WHO 2017)	New Evidence Impact	Updated Recommendation (based on evidence up to 29 April 2026)
Enhanced antimicrobial prophylaxis (cotrimoxazole + isoniazid preventive therapy + fluconazole pre-emptive therapy for CrAg-positive) should be considered in high-burden settings (Conditional recommendation, moderate-quality evidence).	The REALITY trial's 5-year results demonstrate sustained survival benefit with the enhanced package, meeting the pre-specified mortality reduction target. No late toxicities or antimicrobial resistance signals emerged, though fluconazole resistance surveillance is still limited.	**Upgrade to strong recommendation, high-quality evidence:** A package of enhanced antimicrobial prophylaxis consisting of cotrimoxazole, isoniazid preventive therapy (after ruling out active TB), and pre-emptive fluconazole for CrAg-positive individuals is strongly recommended for all AHD patients in settings with high TB and cryptococcal burden.

All GRADE assessments in this report follow the GRADE Working Group criteria (quality of evidence: High—further research is very unlikely to change confidence in the estimate of effect; Moderate—further research is likely to have an important impact; Low—further research is very likely to have an important impact; Very Low—any estimate of effect is very uncertain. Strength of recommendation: Strong—desirable effects clearly outweigh undesirable effects or vice versa; Conditional—trade-offs are less certain or confidence in effect estimates is lower).